Analysis of the dataset by comparing the performance of all three models.

After analyzing all three models, it shows that the K-NN model go beyond the evaluation metrics and gives the better result by displaying the highest values in accuracy, precision, recall and F1 score. However if rapidity are significant, SVM and LDA could still be utilized depending on particular requirements.

NAME	k-NN Model performance:	SVM Model performance:	LDA Model performance:
ACCURACY	0.6136	0.5653	0.5306
PRECISION	0.6157	0.5502	0.5102
RECALL	0.6136	0.5653	0.5306
F1-SCORE	0.6140	0.5293	0.5070

Strengths and Weaknesses of K-NN:-

Strengths:-

1) K Nearest Neighbor for this data is 1.
2) Preforms relatively good when there are labelled data points.
3) This model is the best among the three other model.

Weaknesses:-

1) It is delicate to unrelated structures.
2) Comparatively costly for bigger datasets.

Strengths and Weaknesses of SVM:-

Strengths:-

1) The best kernel for this dataset is ‘rbf’.
2) It can handle non-linear data when an exact kernel is used.
3) SVM does it job very efficiently for compound relations.

Weaknesses:-

1) This is not suitable for larger datasets.
2) All the parameters should be handled carefully.
3) Time consuming process.

Strengths and Weaknesses of LDA:-

Strengths:-

1) It is simple and less time consuming.
2) Gives efficient output for probabilistic data.

Weaknesses:-

1) It is sensitive for covariance estimation.
2) It has poor performance compared to K-NN and SVM.